Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

14.1 Efficiency and Equity

- In a market economy, resource allocation under a perfectly competitive market situation is efficient.
- Equity is about the fairness of distributing income or output.
- Whether efficiency is achieved or not, benefits may be unevenly distributed among members of society in all types of economy.
- Efficiency can be measured objectively while equity involves value judgements.

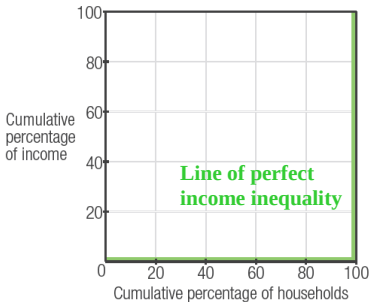
14.2 Measuring Income Inequality

Income distribution

- **Income distribution** can be presented in two ways:
  - By monthly individual or household income
  - By decile group

Lorenz curve

- Meaning of the **Lorenz curve**
  - The Lorenz curve expresses the relationship between the cumulative percentage of households and the cumulative percentage of income.
  - Every point on a Lorenz curve can be interpreted as 'the bottom X% of households earn Y% of the total income'.
- Two extreme cases of the Lorenz curve

Perfect income equality	Perfect income inequality
	
Every household earns the same amount of income.	Only one household earns an income and all the other households earn nothing.

A Lorenz curve never lies above the line of perfect income equality.

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- Comparison between Lorenz curves
  - In reality, the Lorenz curve always lies somewhere between the line of perfect income equality and the line of perfect income inequality.
  - When the income of an economy is distributed more evenly (unevenly), the Lorenz curve will be closer to (further away from) the line of perfect income equality.

Gini coefficient

- Meaning of the **Gini coefficient**
  - The Gini coefficient is an indicator of the degree of income inequality and is often used in conjunction with the Lorenz curve.

$$\text{Gini coefficient} = \frac{\text{Area between the line of perfect income equality and the Lorenz curve}}{\text{Area under the line of perfect income equality}}$$

- Even if the Gini coefficients of two countries are the same, their Lorenz curves and income distributions may be different.
- A higher Gini coefficient does not necessarily imply that the low-income groups have become poorer.

- Three cases of the Gini coefficient

Gini coefficient = 0	0 < Gini coefficient < 1	Gini coefficient = 1
The Lorenz curve coincides with the line of perfect income equality.	• The Lorenz curve lies between the line of perfect income equality and the line of perfect income inequality. • The more even the income distribution, the smaller the Gini coefficient.	The Lorenz curve coincides with the line of perfect income inequality.

Limitations of income inequality measurements

- Different age structures
- Impact of taxes and transfers
- Different household sizes
- Different definitions of income

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14.3 Sources of Income Inequality

- Income from providing human effort (wage income)
  - Difference in natural abilities
  - Difference in human capital
  - Difference in working hours
  - Compensating differentials
  - Wage discrimination
  - Trade union protection
- Income from holding assets (capital income)
  - There is **unequal ownership of assets** such as shares, deposits and properties.
- Government taxes and transfers
  - **Taxes**
    - If a government imposes taxes that are progressive (regressive), the degree of income inequality tends to decrease (increase).
  - **Transfers**
    - If transfers are provided in cash, income of the recipients (usually the low-income groups) will increase, and the post-tax post-social transfer income of the recipients will be higher.
    - If income inequality measurements only reflect inequality in households' monetary income, and transfers are made in kind, the problem of income inequality may be overstated.

14.4 Two Principles of Equity

Equalising income

- **Equalising income** is an idea of fairness which seeks to reduce or eliminate differences in income among individuals or households in an economy.
- Policies for equalising income
  - Taxes
    - A government can impose taxes on high-income groups or adjust the progressiveness of a tax system.
    - **Tax payers (mainly high-income groups) would then have to pay more taxes while non-tax payers (mainly low-income groups) would not be affected.**
  - Transfers
  - Price control, e.g. rent control and the minimum wage law



A minimum wage law may not reduce income inequality as it may push the unemployment rate up.

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- Disincentive effects of taxes and transfers
  - Disincentive on working
    - Taxes like salaries tax weaken the incentive to work harder.
    - Transfers like unemployment benefits weaken the incentive to be employed.
  - Disincentive on saving
    - Taxes like the estate duty discourage saving.
  - Disincentive on investment
    - Taxes like profits tax discourage investment.

Equalising opportunities

- **Equalising opportunities** is an idea of fairness that ensures different people in similar situations are treated similarly.
- Policies for equalising opportunities
  - Policies to eliminate discrimination
    - Ensuring that everyone has an equal opportunity to use his or her talents to earn income.
  - Education policies
    - Ensuring that people with disadvantaged backgrounds **will not be deprived of education because of their backgrounds**.
    - Receiving education will help **improve their productivity** and therefore their **income-earning power in the future**.

14.5 Trade-off between Equity and Efficiency

- Is there a trade-off between equity and efficiency?
  - Giving up efficiency in exchange for equity: Policies that redistribute income from the rich to the poor may lower the working incentive for both low-income and high-income groups.
  - Giving up equity in exchange for efficiency: Policies that encourage competition in the market may worsen income inequality.
- How may redistribution of income improve both efficiency and equity?
  - Policies that create jobs and policies that eliminate discrimination may improve both efficiency and equity.
- How much trade-off is desirable?
  - Economics alone is not enough to determine the balance between equity and efficiency.
  - Nevertheless, economics helps avoid policies that reduce efficiency without improving equity.

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